Fiscal Year:	FY 2020	**Task Last Updated:**	FY 07/03/2020
PI Name:	Cheng, Zhengdong Ph.D.		
Project Title:	Liquid Crystals of Nanoplates		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	zcheng@tamu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	979-845-3413
Organization Name:	Texas A&M University		
PI Address 1:	Chemical Engineering		
PI Address 2:	220 Jack Brown Building, 3122 TAMU		
PI Web Page:			
City:	College Station	**State:**	TX
Zip Code:	77843-0001	**Congressional District:**	17
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	09/01/2013	**End Date:**	08/31/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Tin, Padetha	**Contact Phone:**	216-433-8164
Contact Email:	Padetha.Tin@grc.nasa.gov		
Flight Program:			
Flight Assignment:	Liquid Crystal Facility NOTE: Extended to 8/31/2020 per NSSC information (Ed., 2/14/19)		
Key Personnel Changes/Previous PI:	July 2020 report: Dr. Qingsheng Wang is now Co-PI.		
COI Name (Institution):	Wang, Qingsheng Ph.D. (Co-PI: Texas A&M University)		
Grant/Contract No.:	NNX13AQ60G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The central objectives of the proposed research are to investigate the dynamics of phase transitions, including the formation of liquid crystals as well as the equilibrium gel or empty liquid state, of highly anisotropic colloidal disks. For the FY2020, ground based research was focused on the field induced control of liquid crystal alignment, the effect of temperature gradient on the nucleation, growth, and dispersion of crystals within a channel, as well as investigation of a laser streaming device to induce mixing of samples in space.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	The understanding of the structure and properties of colloidal disks are important to soil science and space exploration (e.g., Mars), to the fundamental investigation of discotic liquid crystals, as well as various industries, including oil recovery, the delivery of pharmaceuticals, nutrition, and cosmetics. The ability to affect the alignment of nanoparticles using external stimuli such as electric field and temperature gradient opens the door to controlled structural assembly of these materials in real-life applications. In this reporting period, we have demonstrated for the first time, to the best of our knowledge, a second order paranematic-biaxial nematic transition in 2D colloidal liquid crystals with negative polarizability. We also experimentally deduced an unprecedented comprehensive phase diagram in an electric field-concentration plane.Through ground experiments, we observe that polarized microscopic observations are most effective at thicknesses around 120 microns, and propose to conduct space microscope experiments studying nucleation and growth at this thickness.
Task Progress:	The phase behavior of nanoparticles is largely dependent on the composition and concentration of particles in the system itself. Furthermore, phase behavior is largely affected by external field such as temperature, magnetic fields, electric fields, and flow fields. External fields have yielded unique mesophase behaviors, as well as better means of control on the dynamics of phase transition and nanoparticle alignment, otherwise conventionally elusive. Herein, we focus on liquid crystal phase transitions and the effects of external electric field and temperature gradient. We explored the details of the 1st order versus the 2nd order transition in discotic liquid crystals in the presence of an external electric field. To the best of our knowledge, this is the first set of studies of this nature, which observes a second order transition in 2D colloidal nanoplates with negative polarizability as well as a comprehensive delineation of the phase diagram. We synthesized Zirconium Phosphate nanoplatelets of negative anisotropic susceptibility by hydrothermal method in volume fractions ranging from 0.003 to 0.015. Under no field, we observe an isotropic to a uniaxial nematic transition as concentration increases. In the presence of an electric field, we experimentally demonstrated the formation of the biaxial nematic phase. The formation of the biaxial nematic phase can be characterized under polarized microscope observations of schlieren textures as splitting of ±1 defects and the re-alignment of domains into higher order with ±1/2 defects. Collating the respective transition points, a phase diagram in the electric field strength - concentration plane can be constructed which reveals the existence of a tricritical point. This tricritical point occurred at the point of convergence between a second order transition and a first order transition. Our methods provides a phase diagram that can be scaled to other 2D materials such as graphene oxide based on particle polarizability and size. Furthermore, we investigated the effect of sample thickness on the isotropic-nematic transition. We observe than in gravity-affected experiments, thicker channels requiring larger volume of material could lead to increased particle sedimentation and distortions in the clarity of microscopic observations. Whilst this observation might be minimized in microgravity, another factor that could be is accumulation of juxtaposed nematic domains/defects. To prevent such complications in-flight, we propose that all flight experiments involving observation of the kinetics be conducted at a cavity thickness around 120 μm. Current /future studies are focused on the potential effectiveness of a laser streaming method to induce mixing of samples during experiments to be conducted in space. Furthermore, we are interested in investigating the combined role of simultaneous magnetic and electric fields on nanoplate alignment and ordering. Finally, we aim to perform temperature gradient induced nematic formation in microfluidic channels of even smaller sizes.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Articles in Peer-reviewed Journals	Shinde A, Huang D, Saldivar M, Xu H, Zeng M, Okeibunor U, Wang L, Mejia C, Tin P, George S, Zhang L, Cheng Z. "Growth of colloidal nanoplate liquid crystals using temperature gradients." ACS Nano. 2019 Nov 26;13(11):12461-9. https://doi.org/10.1021/acsnano.9b01573 , Nov-2019
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Dissertations and Theses	Okeibunor U. (Ugochukwu Okeibunor) "Electric Field Induced Isotropic-Nematic Phase Transitions in Negatively Anisotropic Discotic Liquid Crystals." Texas A&M University, February 2020. , Feb-2020
Dissertations and Theses	Nava I. (Ilse Nava) "Iron-catalyzed Belousov-Zhabotinsky (BZ) hydrogels and liquid crystals." Texas A&M University, June 2020. , Jun-2020